Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

DevOps Engineering on AWS

TGS Ref No: TGS-2023040474

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 14

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
7	31 July 2026	Previous release — 5-day DevOps Engineering on AWS lesson plan.	Dr Alfred Ang
8	23 August 2026	Rebuilt from a single content source aligned to the six official domains and six connected hands-on labs (Labs 1–6); TSC (ICT-DES-4006-1.1) K1–K7/A1–A6/LO1–LO6 mapping added.	Dr Alfred Ang
9	23 August 2026	Aligned to the corrected v9 courseware: concept-led slide deck expanded to 210–230 substantive slides and each lab expanded to 12–16 explicit steps in the Learner Guide/READMEs. Five-day structure unchanged; GitHub via AWS CodeConnections is the primary source path (CodeCommit optional). Full K1–K7/A1–A6/LO1–LO6 mapping preserved.	Dr Alfred Ang
10	23 August 2026	Redesigned to exactly 25 hands-on labs mapped 5 per day across the 5 training days (Day 1 Labs 1–5; Day 2 Labs 6–10; Day 3 Labs 11–15;	Dr Alfred Ang

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		Day 4 Labs 16–20; Day 5 Labs 21–25, including the integrated capstone Lab 25). Daily schedule re-sequenced so each day runs its 5 labs while keeping the five-day structure. All 25 labs mapped across LO1–LO6, K1–K7, A1–A6 and the six official domains; aligned to the v10 slide deck, Learner Guide and lab READMEs. GitHub via AWS CodeConnections remains the primary source path (CodeCommit optional).	
11	23 August 2026	Corrected assessment and delivery timing to match the approved Assessment Plan (v5.0) and Course Proposal: Written Assessment (SAQ) 1 hour + Practical Performance (PP) 2 hours = 3 assessment hours (the PP duration and the assessment total were previously understated). Total delivery re-stated as 40 hours = 37 instructional + 3 assessment: Days 1–4 remain 8 instructional hours each (32h) and Day 5 is 5 instructional hours plus the 3-	Dr Alfred Ang

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		hour assessment. Day 5 schedule re-sequenced to fit all five labs (Labs 21–25) into its 5 instructional hours. All content, 25 labs (5 per day) and visual design unchanged.	
12	23 August 2026	Final-QA corrective release. Daily schedule aligned to the approved 9:30am–6:00pm timing (30-minute lunch; tea breaks within training time) — Days 1–4 remain 8 instructional hours each and Day 5 5 instructional + 3 assessment hours. Added an explicit Slides column to every day table mapping each teaching block/lab to its verified v12 deck slide numbers (read from the deck's slide map). Body text set to Arial 11pt minimum throughout (no sub-11pt overrides). Table of Contents is a live Word TOC field. Content, 25 labs (5 per day) and visual design unchanged.	Dr Alfred Ang
13	23 August 2026	Expanded the five-day trainer deck from 231 to 589 substantive, concept-led slides by blending current courseware with vetted coverage	Dr Alfred Ang

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		from the legacy 583-slide deck and the related AWS DevOps certification course. The Slides column is regenerated from the v13 slide map. The 25-lab plan remains exactly five labs per day; detailed procedures remain in the Learner Guide and lab folders. No practice exam was added.	
14	23 August 2026	Expanded the five-day trainer deck to 629 substantive, concept-led slides by adding 40 vetted AWS DevOps service-boundary architecture maps. The Slides column is regenerated from the v14 slide map. The 25-lab plan remains exactly five labs per day; detailed procedures remain in the Learner Guide and lab folders. No practice exam was added.	Dr Alfred Ang

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Course Information

Course Title	DevOps Engineering on AWS
WSQ Course Reference	TGS-2023040474
Skills Framework (TSC)	Solution Architecture (ICT-DES-4006-1.1) — K1–K7, A1–A6, LO1–LO6
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	5 days · 40 total hours (37 instructional + 3 assessment) · Days 1–4: 8 instructional hours each; Day 5: 5 instructional hours + 3 assessment hours · 25 hands-on labs (5 per day)
Daily Timing	9:30 am – 6:00 pm (30-minute lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the AWS cloud
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Develop an AWS CI/CD pipeline proof of concept.
- LO2: Identify the technical and practical requirements in an AWS cloud infrastructure setup.
- LO3: Develop a resilient cloud solution based on AWS cloud components.
- LO4: Implement regular AWS monitoring and perform necessary troubleshooting.
- LO5: Demonstrate solutions to incidents and business problems on the AWS cloud.
- LO6: Prepare a technical blueprint to address security and compliance.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on AWS DevOps tasks, 2 hours, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The 3 assessment hours (WA 1 hour + PP 2 hours) are conducted after the training content on Day 5.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Daily timing 9:30 am – 6:00 pm (30-minute lunch; tea breaks within training time). The Slides column maps each teaching block or lab to its slide number(s) in the current v14 deck.

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Day 1 — Foundations and SDLC Automation

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory digital attendance (AM)	Slides 2–11
10:00–11:15	75 min	**Topic 1 — SDLC Automation: DevOps culture/CALMS, CI vs CD, the AWS developer tools (source via GitHub + AWS CodeConnections as the primary path, CodeCommit as an AWS-native option; CodeBuild/CodeDeploy/CodePipeline), buildspec & appspec (concepts + demo)**	Slides 16–142
11:15–11:30	15 min	Tea break	—
11:30–13:30	120 min	Hands-on (Lab 1: Set Up a Safe AWS & Git Working Environment and Verify Identity; Lab 2: Establish a Source-Control Workflow and Branching for the Sample App): set up a safe least-privilege AWS/Git workspace and verify identity, then establish the source workflow and branching via GitHub + AWS CodeConnections	Slides 143, 144
13:30–14:00	30 min	Lunch break	—
14:00–15:30	90 min	Deployment	Slide 145

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		strategies (in-place/rolling/blue-green/canary) concepts. Hands-on (Lab 3: Run Automated Tests on Every Commit with AWS CodeBuild): run automated tests in CodeBuild and verify the build gate	
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Hands-on (Lab 4: Produce and Store Versioned Build Artifacts with a Buildspec; Lab 5: Assemble a CodePipeline with Manual Approval and a Deployment Strategy): produce versioned artifacts with buildspec, then assemble CodePipeline with a manual approval and a deployment strategy end to end	Slides 146, 147
17:45–18:00	15 min	Day 1 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 2 — Configuration Management and IaC

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)	—
9:45–11:15	90 min	**Topic 2 — Configuration Management and IaC: IaC principles, CloudFormation vs CDK vs Terraform, template anatomy, change sets &**	Slides 149–243

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		drift, Systems Manager, multi-account/Region (concepts + demo)	
11:15–11:30	15 min	Tea break	—
11:30–13:30	120 min	Hands-on (Lab 6: Deploy a Reusable VPC Network Baseline with CloudFormation; Lab 7: Deploy a Parameterized Application Stack that Imports Network Exports): deploy the CloudFormation network baseline, then a parameterised application stack that imports its exports	Slides 244, 245
13:30–14:00	30 min	Lunch break	—
14:00–15:30	90 min	Hands-on (Lab 8: Preview Changes with Change Sets and Detect & Reconcile Drift; Lab 9: Externalize Configuration with SSM Parameter Store): preview changes with change sets and detect drift; externalise configuration with Systems Manager Parameter Store	Slides 246, 247
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Hands-on (Lab 10: Multi-Account, Multi-Region Guardrails with CloudFormation StackSets): build a multi-account / multi-Region	Slide 248

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		deployment blueprint with CloudFormation StackSets and AWS Organizations; tear down cleanly	
17:45–18:00	15 min	Day 2 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 3 — Resilient Cloud Solutions

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)	—
9:45–11:15	90 min	**Topic 3 — Resilient Cloud Solutions: high availability, Auto Scaling, load balancer types, decoupling, RTO/RPO and DR patterns, backup & self-healing (concepts + demo)**	Slides 250–350
11:15–11:30	15 min	Tea break	—
11:30–13:30	120 min	Hands-on (Lab 11: Deploy a Multi-AZ Application Load Balancer with an Auto Scaling Group; Lab 12: Attach Target-Tracking Auto Scaling and Validate Scale-Out Under Load): deploy a multi-AZ ALB + Auto Scaling group, then attach a target-tracking policy and validate scale-out under load	Slides 351, 352
13:30–14:00	30 min	Lunch break	—
14:00–15:30	90 min	Hands-on (Lab 13:	Slide 353

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		Select a DR Strategy by RTO/RPO and Implement a Pilot-Light Skeleton): select a DR pattern against RTO/RPO and implement a pilot-light skeleton in a second Region	
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Hands-on (Lab 14: Automate Backup, Restore and Data Protection with AWS Backup; Lab 15: Inject Failure and Verify Self-Healing with AWS FIS): automate backup/restore and data protection, then inject failure and verify self-healing	Slides 354, 355
17:45–18:00	15 min	Day 3 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 4 — Monitoring, Logging, and Incident Response

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 3 recap and mandatory digital attendance (AM)	—
9:45–11:00	75 min	**Topic 4 — Monitoring and Logging: CloudWatch metrics/logs/dashboards/alarms, the logs-to-action pipeline, X-Ray and Insights (concepts + demo)**	Slides 357–440
11:00–11:15	15 min	Tea break	—
11:15–13:30	135 min	Hands-on (Lab 16: Collect Custom Metrics with the	Slides 441, 442, 443

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		CloudWatch Agent and Build an Operations Dashboard; Lab 17: Centralise Logs in CloudWatch Logs and Analyse Them with Logs Insights; Lab 18: Turn Logs into Signals: Metric Filters, CloudWatch Alarms and SNS Notifications): build CloudWatch metrics & a dashboard, centralise logs and query with Logs Insights, then wire metric filters, alarms and SNS	
13:30–14:00	30 min	Lunch break	—
14:00–15:30	90 min	**Topic 5 — Incident and Event Response: event-driven ops, EventBridge, SNS, Lambda & Systems Manager auto-remediation, runbooks and deployment rollback (concepts + demo)**	Slides 445–522
15:30–15:45	15 min	Tea break	—
15:45–17:45	120 min	Hands-on (Lab 19: Route Incidents with Amazon EventBridge; Lab 20: Automated Remediation with Lambda and Systems Manager): route incidents with EventBridge, then remediate automatically with Lambda / Systems Manager; simulate an incident and	Slides 523, 524

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		verify auto-recovery	
17:45–18:00	15 min	Day 4 recap, Q&A and PM digital attendance	—

Total training time: 480 minutes (8 hours).

Day 5 — Security, Compliance, and Capstone

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 4 recap and mandatory digital attendance (AM)	—
9:45–10:30	45 min	**Topic 6 — Security and Compliance: IAM least privilege, IAM at scale, encryption (KMS/Secrets), CloudTrail, AWS Config, GuardDuty/Security Hub and the compliance blueprint (concepts + demo)**	Slides 526–610
10:30–10:45	15 min	Tea break	—
10:45–12:45	120 min	Hands-on (Lab 21: Enforce IAM Least Privilege and Analyse Policies for Over-Permissive Access; Lab 22: Protect Data with Secrets Manager and Customer-Managed KMS Encryption; Lab 23: Produce CloudTrail Audit Evidence and Answer Who-Did-What-When): enforce and analyse IAM least privilege, protect data with Secrets Manager and KMS encryption, then produce CloudTrail audit evidence	Slides 611, 612, 613

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12:45–13:15	30 min	Lunch break	—
13:15–14:45	90 min	Hands-on (Lab 24: Run Continuous Compliance with AWS Config and Security Hub; Lab 25: Capstone — Blueprint, Well-Architected Review and Evidence Pack for the Whole Delivery System): run continuous compliance with AWS Config and Security Hub (inject a violation, remediate to COMPLIANT), then the integrated capstone — assemble the technical blueprint, run the operational (Well-Architected) review and compile the evidence pack; cross-domain integration review and TRAQOM survey	Slides 614, 615
14:45–15:00	15 min	Briefing for Assessment; assessment digital attendance	Slide 13
15:00–16:00	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**	Slides 14–15
16:00–18:00	120 min	**Practical Performance (PP) — hands-on AWS DevOps tasks, 2 hours, open book. Assessment digital attendance**	Slides 14–15

Total training time: 300 minutes (5 hours). Assessment: 180 minutes (3 hours).

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Lab & Domain Reference (25 labs · 5 per day · aligned to the six official domains)

Domain / skill area	Weighting	LO · K/A	Lab
Domain 01: SDLC Automation	22%	LO1 · K1, A1	Lab 1: Set Up a Safe AWS & Git Working Environment and Verify Identity; Lab 2: Establish a Source-Control Workflow and Branching for the Sample App; Lab 3: Run Automated Tests on Every Commit with AWS CodeBuild; Lab 4: Produce and Store Versioned Build Artifacts with a Buildspec; Lab 5: Assemble a CodePipeline with Manual Approval and a Deployment Strategy
Domain 02: Configuration Management and IaC	17%	LO2 · K3, A3	Lab 6: Deploy a Reusable VPC Network Baseline with CloudFormation; Lab 7: Deploy a Parameterized Application Stack that Imports Network Exports; Lab 8: Preview Changes with Change Sets and Detect & Reconcile Drift; Lab 9: Externalize Configuration with SSM Parameter Store; Lab 10: Multi-Account, Multi-Region Guardrails with CloudFormation

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			StackSets
Domain 03: Resilient Cloud Solutions	15%	LO3 · K2, K4, A2	Lab 11: Deploy a Multi-AZ Application Load Balancer with an Auto Scaling Group; Lab 12: Attach Target-Tracking Auto Scaling and Validate Scale-Out Under Load; Lab 13: Select a DR Strategy by RTO/RPO and Implement a Pilot-Light Skeleton; Lab 14: Automate Backup, Restore and Data Protection with AWS Backup; Lab 15: Inject Failure and Verify Self-Healing with AWS FIS
Domain 04: Monitoring and Logging	15%	LO4 · K5, A6	Lab 16: Collect Custom Metrics with the CloudWatch Agent and Build an Operations Dashboard; Lab 17: Centralise Logs in CloudWatch Logs and Analyse Them with Logs Insights; Lab 18: Turn Logs into Signals: Metric Filters, CloudWatch Alarms and SNS Notifications
Domain 05: Incident and Event Response	14%	LO5 · K7, A5	Lab 19: Route Incidents with Amazon EventBridge; Lab 20: Automated Remediation with Lambda and Systems Manager

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Domain 06: Security and Compliance	17%	LO6 · K6, A4	Lab 21: Enforce IAM Least Privilege and Analyse Policies for Over-Permissive Access; Lab 22: Protect Data with Secrets Manager and Customer-Managed KMS Encryption; Lab 23: Produce CloudTrail Audit Evidence and Answer Who-Did-What-When; Lab 24: Run Continuous Compliance with AWS Config and Security Hub; Lab 25: Capstone — Blueprint, Well-Architected Review and Evidence Pack for the Whole Delivery System

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